#04 AWS Rekognition Image Intelligence Pipeline

AWS / Computer Vision

Scale image analysis to thousands per day using managed AWS computer vision

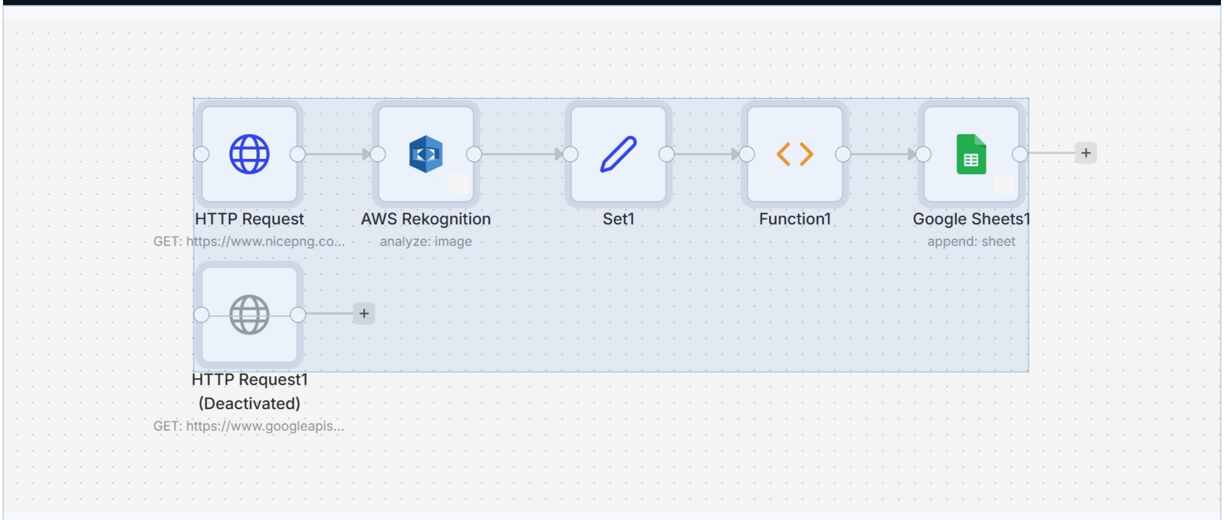
BUSINESS PROBLEM

Manual image review doesn't scale. This pipeline uses AWS Rekognition to detect objects, faces, and text in images at scale — processing thousands per hour with zero ML model training, logging confidence-scored results and routing by content type automatically.

n8n CANVAS — actual workflow screenshot

n8n WORKFLOW EDITOR

0150_Awsrekognition_GoogleSheets_Automation _Webhook.json



HTTP Request → AWS Rekognition → Set1 → Function1 → Google Sheets

INPUT / OUTPUT — exact n8n JSON format

INPUT — HTTP GET — image URL fetched and sent to Rekognition

```
{
  "image_url":
"https://cdn.example.com/product/cam-feed-4421.jpg",
  "image_id":  "prod-4421",
  "source":    "warehouse-cam-3",
  "timestamp": "2024-06-15T08:14:22Z"
}
```

OUTPUT — Google Sheets row appended — analysis result

```
{
  "json": {
    "image_id":   "prod-4421",
    "labels": ["Cardboard Box","Pallet","Forklift","Person"],
    "confidence": [99.1, 97.4, 91.2, 88.6],
    "text_detected": "LOT-2024-0441",
    "flag":       "AMBER",
    "flag_reason":"Person detected in automated zone",
    "action":     "Alert supervisor — zone safety check required",
    "processing_ms": 340
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ **HTTP Request — fetch image**	Webhook or schedule triggers fetch of image URL. Downloads binary for Rekognition. Sets Content-Type: image/jpeg.
2	■ **AWS Rekognition — DetectLabels**	API call: rekognition:DetectLabels with MaxLabels:20, MinConfidence:70. Returns [{Name, Confidence, BoundingBox}].
3	■ **HTTP Request — DetectText (optional)**	Second Rekognition call: reads lot numbers, serial numbers, warning labels, signage from the image.
4	■ **Set Node — filter and transform**	Keeps labels with Confidence above 80%. Adds image_id, timestamp, source camera. Removes raw API boilerplate.
5	■ **Function Node — business logic**	JS: if label==Person and zone==automated → flag=AMBER + alert. if label==Fire → flag=RED + emergency alert.
6	■ **Google Sheets — append result row**	Row: {image_id, labels[], max_confidence, flag, text_detected, processing_ms, camera_source}.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
AWS Access Key + Secret	AWS Console → IAM → Users → Security Credentials → Create access key	AWS Credentials
AWS Region	Set in node — use me-south-1 (Bahrain) or us-east-1	AWS node config
Google Sheets OAuth2	console.cloud.google.com → Sheets API	Google Sheets OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A logistics warehouse uses 24 fixed cameras to monitor loading bays. Images are captured every 30 seconds and sent to this workflow for automated safety and inventory monitoring.
Input received	Camera 3 image at 08:14. Warehouse management system posted image URL to webhook. Image shows a loading bay with pallets and a forklift in motion.
What the workflow found	Rekognition returned: Cardboard Box (99.1%), Pallet (97.4%), Forklift (91.2%), Person (88.6%). Text scan found lot label: LOT-2024-0441. Person in automated forklift zone triggered AMBER flag.
Output produced	Google Sheets row appended. Supervisor notified via Slack: 'Person detected in automated zone — camera 3, 08:14:22.' Zone was cleared and confirmed safe at 08:17. No incident logged.

JSON: 0150_Awsrekognition_GoogleSheets_Automation_Webhook.json	**GitHub: trm-aws-rekognition-pipeline**	aws.amazon.com/rekognition/